

# LAPORAN PENGUJIAN KUALITAS PERANGKAT LUNAK

## INFORMASI PROYEK

Nama Proyek: [Nama Aplikasi/Sistem]

Versi: [v1.0.0]

Tanggal Pengujian: [DD/MM/YYYY - DD/MM/YYYY]

Disusun Oleh: [Nama Tim QA]

Departemen: [Quality Assurance]

## 1. RINGKASAN EKSEKUTIF

### 1.1 Tujuan Pengujian

Dokumen ini menyajikan hasil pengujian kualitas perangkat lunak secara komprehensif meliputi automated testing, load testing, dan User Acceptance Testing (UAT) untuk memastikan sistem memenuhi standar kualitas yang ditetapkan sebelum diluncurkan ke production.

### 1.2 Ringkasan Hasil

- Status Keseluruhan: [PASS/FAIL/CONDITIONAL PASS]
- Total Test Cases: [XXX]
- Pass Rate: [XX%]
- Critical Issues: [X]
- Rekomendasi: [LANJUT/TUNDA/PERBAIKAN DIPERLUKAN]

## 2. CAKUPAN PENGUJIAN

### 2.1 Fitur yang Diuji

- Modul Login dan Autentikasi
- Manajemen Data User
- Dashboard dan Reporting
- Proses Transaksi
- Integrasi dengan Sistem Eksternal

### 2.2 Fitur yang Tidak Diuji

- [Fitur X - Alasan: Belum dikembangkan]
- [Fitur Y - Alasan: Out of scope]

### 2.3 Lingkungan Pengujian

- Environment: Staging/Pre-Production
- URL: <https://staging.example.com>
- Database: PostgreSQL 14.5

- **Server:** AWS EC2 t3.medium
  - **Browser:** Chrome 120, Firefox 121, Safari 17
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### 3. AUTOMATED TESTING

#### 3.1 Unit Testing

##### 3.1.1 Ringkasan Eksekusi

- **Framework:** Jest/JUnit/PyTest
- **Total Test Cases:** 450
- **Passed:** 442
- **Failed:** 8
- **Success Rate:** 98.2%
- **Durasi Eksekusi:** 2 menit 34 detik

##### 3.1.2 Code Coverage

- **Line Coverage:** 87%
- **Branch Coverage:** 82%
- **Function Coverage:** 91%
- **Statement Coverage:** 88%

##### 3.1.3 Hasil Detail

| Modul           | Total Tests | Passed | Failed | Coverage |
|-----------------|-------------|--------|--------|----------|
| Authentication  | 85          | 85     | 0      | 94%      |
| User Management | 120         | 118    | 2      | 89%      |
| Payment Process | 95          | 92     | 3      | 85%      |
| Reporting       | 75          | 73     | 2      | 82%      |
| API Integration | 75          | 74     | 1      | 88%      |

##### 3.1.4 Bug yang Ditemukan

1. **Bug ID: UT-001 - CRITICAL**
  - Deskripsi: Payment calculation error ketika discount > 50%
  - Status: Fixed
  - Test Case: `test_payment_with_high_discount()`
2. **Bug ID: UT-002 - HIGH**
  - Deskripsi: User role validation gagal untuk role 'admin'
  - Status: Open
  - Test Case: `test_admin_role_validation()`

#### 3.2 Integration Testing

##### 3.2.1 Ringkasan Eksekusi

- **Framework:** Selenium/Cypress/Postman
- **Total Test Cases:** 125
- **Passed:** 118

- **Failed:** 7
- **Success Rate:** 94.4%

### 3.2.2 API Testing Results

| Endpoint              | Method | Status | Response Time | Result |
|-----------------------|--------|--------|---------------|--------|
| /api/auth/login       | POST   | 200    | 145ms         | PASS   |
| /api/users            | GET    | 200    | 89ms          | PASS   |
| /api/users/{id}       | PUT    | 200    | 234ms         | PASS   |
| /api/transactions     | POST   | 500    | -             | FAIL   |
| /api/reports/generate | GET    | 200    | 1.2s          | PASS   |

### 3.2.3 Bug yang Ditemukan

1. **Bug ID: IT-001 - CRITICAL**
  - Deskripsi: Transaction API gagal saat concurrent request > 10
  - Status: In Progress
  - Impact: High volume transactions akan gagal

## 3.3 End-to-End Testing

### 3.3.1 Ringkasan Eksekusi

- **Framework:** Selenium/Playwright
- **Total Scenarios:** 45
- **Passed:** 41
- **Failed:** 4
- **Success Rate:** 91.1%

### 3.3.2 Test Scenarios

| Scenario                | Steps | Result | Screenshot |
|-------------------------|-------|--------|------------|
| User Registration Flow  | 8     | PASS   | ✓          |
| Login to Dashboard      | 5     | PASS   | ✓          |
| Complete Purchase       | 12    | FAIL   | ✓          |
| Generate Monthly Report | 6     | PASS   | ✓          |

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## 4. LOAD TESTING

### 4.1 Tujuan dan Metodologi

Menguji performa sistem di bawah beban tinggi untuk memastikan sistem dapat menangani traffic yang diharapkan.

- **Tool:** Apache JMeter / K6 / Gatling
- **Durasi Test:** 60 menit
- **Ramp-up Period:** 10 menit

### 4.2 Skenario Load Testing

#### Skenario 1: Normal Load

- **Virtual Users:** 100

- **Duration:** 30 menit
- **Transactions/sec:** 50 TPS

#### Skenario 2: Peak Load

- **Virtual Users:** 500
- **Duration:** 20 menit
- **Transactions/sec:** 200 TPS

#### Skenario 3: Stress Test

- **Virtual Users:** 1000
- **Duration:** 10 menit
- **Transactions/sec:** 400 TPS

### 4.3 Hasil Performance Testing

#### 4.3.1 Response Time

| Metric                | Normal Load | Peak Load | Stress Test | SLA Target |
|-----------------------|-------------|-----------|-------------|------------|
| Average Response Time | 245ms       | 850ms     | 2.1s        | < 1s       |
| 90th Percentile       | 450ms       | 1.5s      | 4.2s        | < 2s       |
| 95th Percentile       | 680ms       | 2.3s      | 6.8s        | < 3s       |
| Max Response Time     | 1.2s        | 5.4s      | 12.3s       | < 5s       |

#### 4.3.2 Throughput

| Load Scenario | Requests/sec | Successful | Failed | Error Rate |
|---------------|--------------|------------|--------|------------|
| Normal Load   | 52           | 93,600     | 0      | 0%         |
| Peak Load     | 198          | 237,600    | 1,200  | 0.5%       |
| Stress Test   | 385          | 231,000    | 28,800 | 11.1%      |

#### 4.3.3 Resource Utilization

| Resource             | Normal Load | Peak Load | Stress Test | Threshold  |
|----------------------|-------------|-----------|-------------|------------|
| CPU Usage            | 45%         | 78%       | 95%         | < 80%      |
| Memory Usage         | 3.2 GB      | 5.8 GB    | 7.1 GB      | < 6 GB     |
| Network I/O          | 25 Mbps     | 89 Mbps   | 156 Mbps    | < 100 Mbps |
| Database Connections | 45          | 185       | 298         | < 200      |

### 4.4 Bottleneck Analysis

#### 1. Database Query Performance

- Issue: Query `/api/reports` membutuhkan waktu 3.5s pada peak load
- Root Cause: Missing index pada table transactions
- Recommendation: Tambahkan composite index

#### 2. Connection Pool Exhaustion

- Issue: Connection timeout pada 1000 concurrent users
- Root Cause: Max pool size = 200
- Recommendation: Increase pool size ke 500

#### 3. Memory Leak

- Issue: Memory usage terus meningkat seiring waktu
- Root Cause: Session objects tidak di-cleanup
- Recommendation: Implement proper session management

### 4.5 Kesimpulan Load Testing

- ✓ Sistem stabil pada normal load (100 users)
- ⚠ Performance degradation signifikan pada peak load (500 users)
- ✗ Sistem tidak stabil pada stress test (1000 users)
- **Rekomendasi:** Optimisasi database dan increase server capacity sebelum production

## 5. USER ACCEPTANCE TESTING (UAT)

### 5.1 Informasi UAT

- **Periode UAT:** 15 Januari 2025 - 29 Januari 2025
- **Jumlah Tester:** 15 user representatives
- **Environment:** UAT Environment (production-like)
- **Metode:** Black-box testing berdasarkan business scenarios

### 5.2 Partisipan UAT

| No | Nama           | Departemen       | Role       | Status      |
|----|----------------|------------------|------------|-------------|
| 1  | Ahmad Wijaya   | Sales            | End User   | Completed   |
| 2  | Siti Nurhaliza | Finance          | Power User | Completed   |
| 3  | Budi Santoso   | Operations       | Manager    | In Progress |
| 4  | Dewi Lestari   | Customer Service | End User   | Completed   |
| 5  | Eko Prasetyo   | IT               | Admin      | Completed   |

### 5.3 Business Scenarios

#### Scenario 1: Proses Penjualan End-to-End

**Tester:** Ahmad Wijaya (Sales)

**Status:** APPROVED

**Steps:**

1. Login ke sistem
2. Buat customer baru
3. Input order penjualan
4. Pilih produk dan quantity
5. Apply discount
6. Generate invoice
7. Process payment
8. Print receipt

**Result:** ✓ PASS

**Comments:** "Proses lancar, UI intuitif, response time bagus"

**Issues Found:**

- UAT-001 (LOW): Tombol print tidak responsive di Safari browser

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### Scenario 2: Laporan Keuangan Bulanan

**Tester:** Siti Nurhaliza (Finance)

**Status:** APPROVED WITH CONDITIONS

**Steps:**

1. Login sebagai finance user
2. Navigate ke menu Reports
3. Pilih periode bulan lalu
4. Filter by department
5. Export ke Excel
6. Verify data accuracy

**Result:**  $\Delta$  PASS WITH CONDITIONS

**Comments:** "Data akurat tapi export Excel lambat untuk data > 10,000 rows"

**Issues Found:**

- UAT-002 (MEDIUM): Export Excel timeout untuk dataset besar
- UAT-003 (LOW): Format tanggal tidak sesuai locale Indonesia

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### Scenario 3: User Management dan Role Assignment

**Tester:** Eko Prasetyo (IT Admin)

**Status:** REJECTED

**Steps:**

1. Login sebagai admin
2. Create new user
3. Assign multiple roles
4. Set department and permissions
5. Send activation email
6. Verify user can login

**Result:** X FAIL

**Comments:** "Critical bug: tidak bisa assign multiple roles sekaligus"

**Issues Found:**

- UAT-004 (CRITICAL): Multiple role assignment gagal
- UAT-005 (HIGH): Activation email tidak terkirim untuk beberapa domain

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## 5.4 Ringkasan Issues UAT

| Priority | Total | Open | In Progress | Resolved | Closed |
|----------|-------|------|-------------|----------|--------|
| CRITICAL | 2     | 1    | 1           | 0        | 0      |

| Priority     | Total     | Open      | In Progress | Resolved | Closed   |
|--------------|-----------|-----------|-------------|----------|----------|
| HIGH         | 5         | 2         | 2           | 1        | 0        |
| MEDIUM       | 8         | 3         | 3           | 2        | 0        |
| LOW          | 12        | 4         | 2           | 4        | 2        |
| <b>TOTAL</b> | <b>27</b> | <b>10</b> | <b>8</b>    | <b>7</b> | <b>2</b> |

## 5.5 User Satisfaction Survey

| Aspek                | Rating (1-5) | Comments                  |
|----------------------|--------------|---------------------------|
| Ease of Use          | 4.2          | Interface cukup intuitif  |
| Performance          | 3.8          | Beberapa fitur lambat     |
| Functionality        | 4.5          | Fitur sesuai kebutuhan    |
| Reliability          | 3.5          | Ada beberapa bug critical |
| Overall Satisfaction | 4.0          | Secara keseluruhan bagus  |

## 5.6 UAT Sign-off

| Stakeholder  | Role               | Decision    | Signature | Date       |
|--------------|--------------------|-------------|-----------|------------|
| Budi Hartono | Business Owner     | APPROVED*   | _____     | DD/MM/YYYY |
| Sarah Wijaya | Head of Operations | CONDITIONAL | _____     | DD/MM/YYYY |
| Agus Suprpto | IT Manager         | REJECTED    | _____     | DD/MM/YYYY |

\*APPROVED dengan catatan: Critical bugs harus diperbaiki sebelum go-live

# 6. DEFECT SUMMARY

## 6.1 Defect Distribution by Severity

| Severity     | Total     | Open      | In Progress | Resolved  | Won't Fix |
|--------------|-----------|-----------|-------------|-----------|-----------|
| CRITICAL     | 5         | 2         | 2           | 1         | 0         |
| HIGH         | 12        | 4         | 5           | 3         | 0         |
| MEDIUM       | 23        | 8         | 9           | 5         | 1         |
| LOW          | 34        | 12        | 8           | 10        | 4         |
| <b>TOTAL</b> | <b>74</b> | <b>26</b> | <b>24</b>   | <b>19</b> | <b>5</b>  |

## 6.2 Defect Distribution by Module

| Module          | Critical | High | Medium | Low | Total |
|-----------------|----------|------|--------|-----|-------|
| Authentication  | 1        | 2    | 3      | 5   | 11    |
| User Management | 2        | 3    | 5      | 8   | 18    |
| Transaction     | 1        | 4    | 7      | 9   | 21    |
| Reporting       | 0        | 2    | 5      | 7   | 14    |
| Integration     | 1        | 1    | 3      | 5   | 10    |

## 6.3 Critical Defects Detail

### BUG-001: Multiple Role Assignment Failure

- **Severity:** CRITICAL
- **Module:** User Management
- **Status:** In Progress

- **Found in:** UAT
- **Description:** Tidak bisa assign lebih dari satu role ke user
- **Impact:** Admin tidak bisa setup user permissions dengan benar
- **Priority:** P0 - Must fix before release

#### BUG-002: Payment Gateway Timeout

- **Severity:** CRITICAL
  - **Module:** Transaction
  - **Status:** Open
  - **Found in:** Load Testing
  - **Description:** Payment gateway timeout pada concurrent transactions > 100
  - **Impact:** Customer tidak bisa complete purchase pada peak hours
  - **Priority:** P0 - Must fix before release
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## 7. TEST METRICS

### 7.1 Test Execution Metrics

- **Total Test Cases Planned:** 620
- **Total Test Cases Executed:** 620
- **Test Cases Passed:** 560 (90.3%)
- **Test Cases Failed:** 60 (9.7%)
- **Test Cases Blocked:** 0
- **Test Execution Rate:** 100%

### 7.2 Defect Detection Metrics

- **Total Defects Found:** 74
- **Defects per Test Case:** 0.12
- **Critical Defects Density:** 0.8%
- **Defect Resolution Rate:** 32.4% (24/74)
- **Average Time to Fix:** 2.3 days

### 7.3 Test Coverage

- **Requirements Coverage:** 95% (190/200 requirements tested)
  - **Code Coverage:** 87%
  - **Business Process Coverage:** 100%
- 

## 8. RISK ASSESSMENT

### 8.1 Technical Risks

| Risk                             | Probability | Impact | Mitigation                          |
|----------------------------------|-------------|--------|-------------------------------------|
| Database performance degradation | HIGH        | HIGH   | Implement caching, optimize queries |
| Third-party API downtime         | MEDIUM      | HIGH   | Implement retry mechanism & circuit |



| Risk                       | Probability | Impact   | Mitigation                            |
|----------------------------|-------------|----------|---------------------------------------|
| Security vulnerabilities   | LOW         | CRITICAL | Conduct security audit before release |
| Data loss during migration | LOW         | CRITICAL | Multiple backup & rollback strategy   |

## 8.2 Business Risks

| Risk                        | Probability | Impact | Mitigation                     |
|-----------------------------|-------------|--------|--------------------------------|
| User adoption resistance    | MEDIUM      | HIGH   | Comprehensive training program |
| Business process disruption | HIGH        | MEDIUM | Phased rollout approach        |
| Competitor advantage        | LOW         | MEDIUM | Fast-track critical features   |

# 9. RECOMMENDATIONS

## 9.1 Technical Recommendations

### 1. Database Optimization (Priority: HIGH)

- Add missing indexes pada tables: transactions, user\_logs
- Implement query caching untuk reporting queries
- Consider read replicas untuk reporting workload

### 2. Performance Improvements (Priority: HIGH)

- Increase server capacity: upgrade ke t3.large
- Implement CDN untuk static assets
- Enable gzip compression

### 3. Code Quality (Priority: MEDIUM)

- Fix all critical dan high severity bugs
- Increase unit test coverage ke minimal 90%
- Implement automated code review tools

### 4. Security Enhancements (Priority: HIGH)

- Conduct penetration testing
- Implement rate limiting pada API endpoints
- Enable 2FA untuk admin accounts

## 9.2 Process Recommendations

### 1. Testing Process

- Integrate automated tests ke CI/CD pipeline
- Implement smoke testing untuk setiap deployment
- Schedule regular regression testing

### 2. Release Strategy

- Implement blue-green deployment
- Conduct limited beta release sebelum full rollout
- Prepare detailed rollback procedures

### 3. Monitoring & Support

- Setup real-time monitoring dengan alerts

- Prepare incident response procedures
- Train support team sebelum go-live

## 10. GO-LIVE READINESS

### 10.1 Checklist

| Category       | Item                         | Status | Notes                |
|----------------|------------------------------|--------|----------------------|
| Testing        | All test cases executed      | ✓      | Complete             |
| Testing        | Critical bugs resolved       | ✗      | 2 open critical bugs |
| Testing        | UAT sign-off obtained        | ⚠      | Conditional approval |
| Infrastructure | Production environment ready | ✓      | -                    |
| Infrastructure | Backup & recovery tested     | ✓      | -                    |
| Documentation  | User manual completed        | ✓      | -                    |
| Documentation  | Technical documentation      | ✓      | -                    |
| Training       | User training completed      | ⚠      | 80% complete         |
| Training       | Support team trained         | ✓      | -                    |
| Security       | Security audit passed        | ✗      | Scheduled next week  |

### 10.2 Go-Live Decision

Status: ⚠ **CONDITIONAL GO-LIVE**

#### Conditions:

1. BUG-001 (Multiple role assignment) MUST be fixed
2. BUG-002 (Payment gateway timeout) MUST be fixed
3. Security audit must be completed and passed
4. Complete remaining user training sessions

## 11. APPENDIX

### 11.1 Test Data

- Test data location: /test-data/
- Database backup: uat\_backup\_20250129.sql
- Test accounts: See document test\_accounts.xlsx

### 11.2 Test Scripts

- Automated test repository: <https://github.com/company/test-automation>
- Load test scripts: /load-tests/jmeter/
- UAT test cases: /uat/test-cases.xlsx

### 11.3 Screenshots & Evidence

- All test execution screenshots: /evidence/screenshots/
- Performance test reports: /evidence/performance/
- Bug reports: Jira project QA-2025

## **11.4 Tools Used**

- Unit Testing: Jest 29.5
- Integration Testing: Cypress 12.0
- Load Testing: Apache JMeter 5.5
- Bug Tracking: Jira
- Test Management: TestRail
- CI/CD: Jenkins